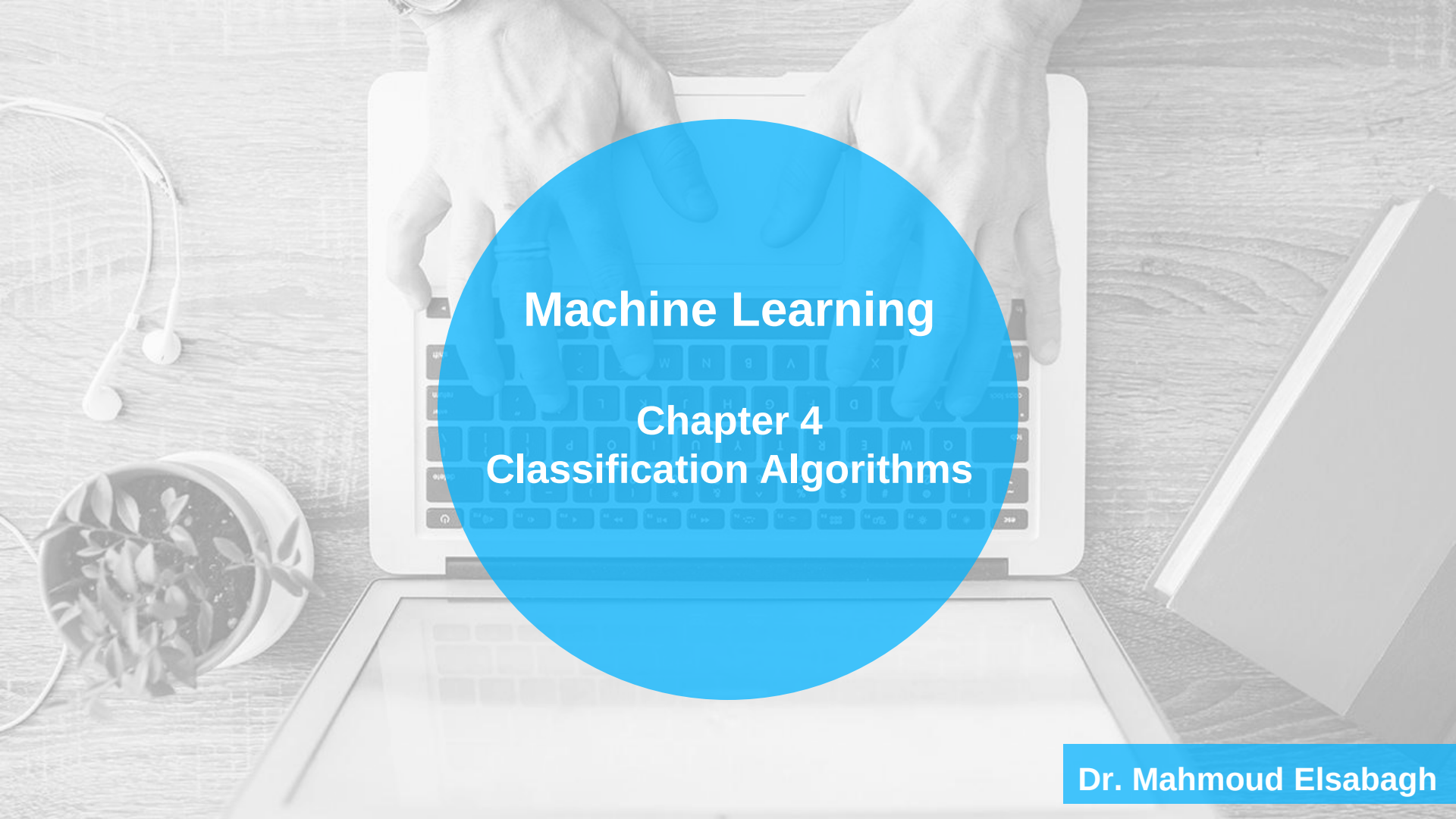




Machine Learning

Chapter 4

Classification Algorithms



Contents

Chapter 1: Introduction to Machine Learning

Chapter 2: Supervised Learning Fundamentals

Chapter 3: Regression Algorithms

Chapter 4: Classification Algorithms

Chapter 5: Decision Trees and Ensemble Learning

Chapter 6: Unsupervised Learning Basics

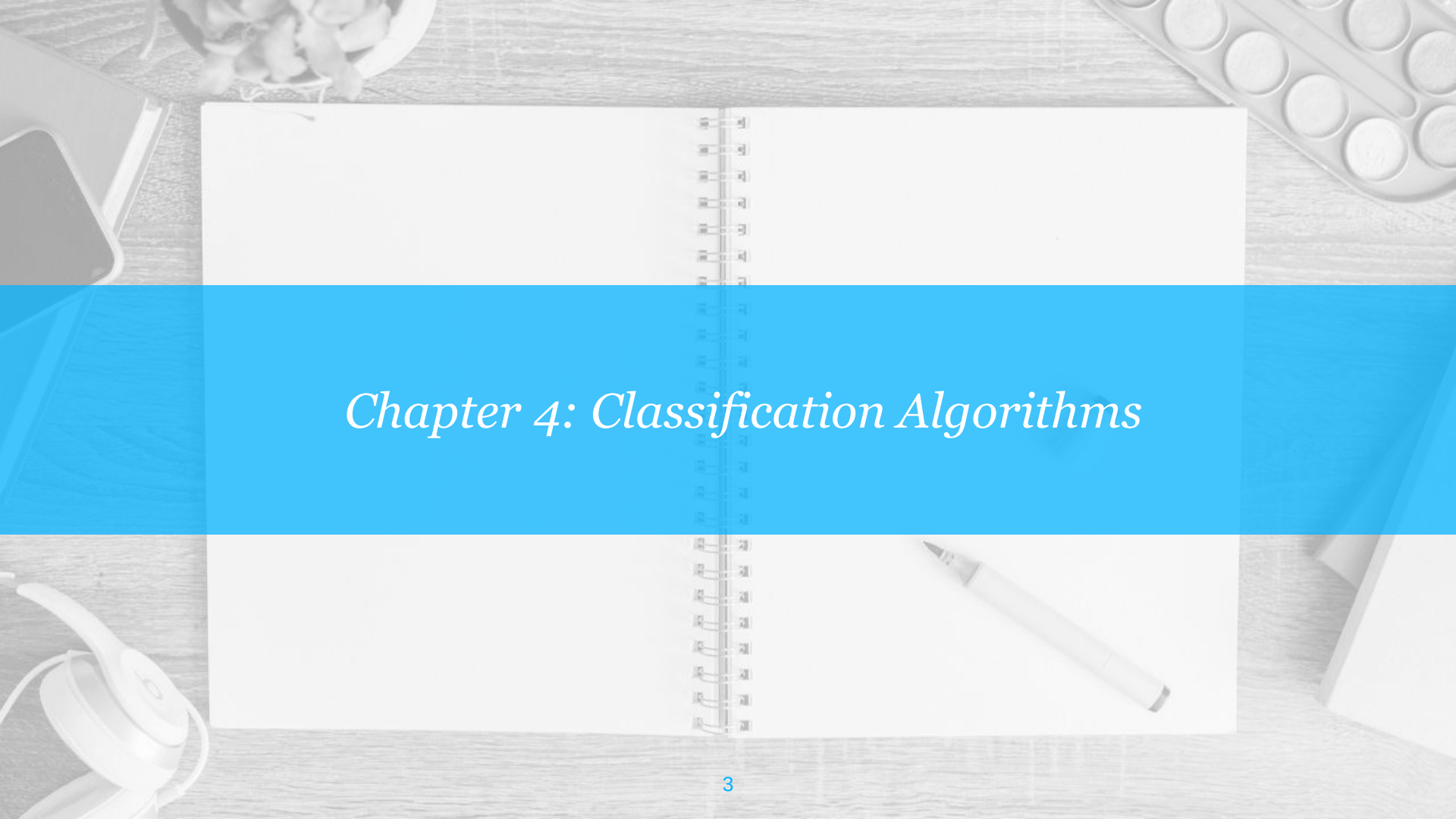
Chapter 7: Neural Networks and Deep Learning

Chapter 8: Natural Language Processing (NLP) and Sequence Models

Chapter 9: Reinforcement Learning

Chapter 10: Model Evaluation and Tuning

Chapter 11: Model Deployment and Applications

A top-down view of a wooden desk. In the center is a spiral-bound notebook with blank white pages. A silver pen lies on the bottom right page. To the top right is a grey paint palette with several wells. To the top left is a small bowl of fruit. A blue horizontal band is overlaid across the middle of the notebook.

Chapter 4: Classification Algorithms



Contents

- 1. Introduction to Classification**
- 2. Types of Classification Algorithms**
- 3. Logistic Regression**
- 4. k-Nearest Neighbors (k-NN)**
- 5. Support Vector Machines (SVM)**
- 6. Decision Tree Classifier**
- 7. Random Forest Classifier**
- 8. Naive Bayes**
- 9. Artificial Neural Networks (ANN)**
- 10. Gradient Boosting Machines (GBM)**
- 11. AdaBoost**
- 12. XGBoost**
- 13. Applications of Classification Algorithms**

1. Introduction to Classification

→ Definition:

Classification is a type of **supervised learning** where the goal is to predict the categorical class labels of new observations based on past observations. The predicted output is discrete, meaning it belongs to a specific class or category.

→ Common Use Cases:

Email classification (spam vs. non-spam).

Disease diagnosis (disease vs. no disease).

Image recognition (dog vs. cat).

Sentiment analysis (positive vs. negative).

2. Types of Classification Algorithms

- Logistic Regression**
- k-Nearest Neighbors (k-NN)**
- Support Vector Machines (SVM)**
- Decision Tree Classifier**
- Random Forest Classifier**
- Naive Bayes**
- Artificial Neural Networks (ANN)**
- Gradient Boosting Machines (GBM)**
- AdaBoost**
- XGBoost**

3. Logistic Regression

→ Concept:

Despite its name, **Logistic Regression** is a classification algorithm. It models the probability that an instance belongs to a specific class using the **sigmoid function**. This algorithm is best suited for **binary classification** tasks (where there are only two classes).

→ Mathematical Representation:

$$P(y = 1|x) = \frac{1}{1 + e^{-z}}$$

→ Decision Boundary:

$$\text{where } z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b.$$

Logistic regression predicts probabilities, but class labels are assigned by applying a threshold (typically 0.5). If the predicted probability is greater than 0.5, the instance is classified as class 1; otherwise, it's classified as class 0.

→ Use Case:

Ideal for binary classification problems such as predicting whether a student will pass or fail based on study hours.

4. k-Nearest Neighbors (k-NN)

→ Concept:

k-NN is a **non-parametric, instance-based learning** algorithm. The idea is simple: to classify a new data point, the algorithm finds the **k** nearest points (neighbors) in the training set and assigns the majority class label among them to the new point.

→ How it Works:

- Choose a value for k (the number of neighbors to consider).
- Compute the distance between the new data point and every point in the training set (commonly using **Euclidean distance**).
- Select the k nearest neighbors.
- Assign the class label that is most common among the k neighbors.

4. k-Nearest Neighbors (k-NN)

→ Advantages:

- Simple and intuitive.
- No need for a training phase (lazy learner).

→ Challenges:

- Computationally expensive for large datasets (since distances must be computed for all points).
- Sensitive to the choice of k and the distance metric used.

5. Support Vector Machines (SVM)

→ Concept:

SVM is a **discriminative classifier** that works by finding the hyperplane that best separates the data into different classes. For **linearly separable data**, SVM finds the hyperplane that maximizes the **margin** between the two classes.

→ Mathematical Representation: The decision boundary is represented by a hyperplane:

$$w \cdot x + b = 0$$

→ Key Feature:

where w is the weight vector and b is the bias.

SVM focuses on the **support vectors**, which are the data points that are closest to the decision boundary. These points are most critical for determining the placement of the hyperplane.

5. Support Vector Machines (SVM)

→ Kernel Trick:

When data is not linearly separable, SVM can use the **kernel trick** to project the data into a higher-dimensional space where a hyperplane can be used to separate the classes. Common kernels include:

Linear Kernel.

Polynomial Kernel.

Radial Basis Function (RBF) Kernel.

→ Advantages:

- Effective in high-dimensional spaces.
- Works well for both linearly and non-linearly separable data (with kernels).

→ Challenges:

Computationally expensive for large datasets.

Sensitive to the choice of kernel and hyperparameters.

6. Decision Tree Classifier

→ Concept:

A decision tree classifier builds a **tree structure** where each node represents a feature and the branches represent the possible values of that feature. The leaves of the tree represent the predicted class labels.

→ How it Works:

- The algorithm recursively splits the dataset into smaller subsets based on the feature that maximizes a certain criterion (e.g., **Gini Impurity** or **Entropy** in case of classification).
- The process continues until the data is perfectly classified or another stopping condition is met (e.g., maximum tree depth).

6. Decision Tree Classifier

→ Advantages:

Simple to understand and interpret.

Can handle both numerical and categorical data.

Non-parametric (no assumptions about the data distribution).

→ Challenges:

Prone to overfitting, especially if the tree is too deep.

Can create biased trees if some classes dominate.

7. Random Forest Classifier

→ Concept:

Random Forest is an **ensemble learning** method that builds multiple decision trees and combines their predictions (via majority voting in classification). The idea is that by aggregating multiple decision trees, Random Forest reduces the risk of overfitting and improves generalization.

→ How it Works:

- Multiple decision trees are trained on different random subsets of the training data.
- The final prediction is made by taking a **majority vote** from all the trees.

→ Advantages:

- Reduces overfitting compared to individual decision trees.
- Works well with large datasets and many features.

→ Challenges:

More complex and less interpretable than a single decision tree.

Requires more computational resources for training.

8. Naive Bayes

→ Concept:

Naive Bayes is based on **Bayes' Theorem** and assumes that features are conditionally independent given the class label (this is the "naive" assumption). Despite this strong assumption, Naive Bayes often performs surprisingly well.

- Bayes' Theorem:

$$P(y|X) = \frac{P(X|y)P(y)}{P(X)}$$

where $P(y|X)$ is the posterior probability of class y given the input features X , $P(X|y)$ is the likelihood, $P(y)$ is the prior, and $P(X)$ is the evidence.

8. Naive Bayes

→ Types of Naive Bayes Classifiers:

Gaussian Naive Bayes: Assumes that the features follow a Gaussian distribution.

Multinomial Naive Bayes: Used for discrete features, often in text classification problems like spam detection.

Bernoulli Naive Bayes: Assumes binary features.

→ Advantages:

Fast to train and predict.

Works well with small datasets and for text classification tasks.

→ Challenges:

Assumes that all features are independent, which is rarely true in real-world data.

13. Applications of Classification Algorithms

- ➔ **Medical Diagnosis:** Predicting the presence or absence of a disease.
- ➔ **Email Filtering:** Classifying emails as spam or not spam.
- ➔ **Customer Segmentation:** Classifying customers based on behavior for targeted marketing.
- ➔ **Image Recognition:** Classifying images into categories (e.g., cats vs. dogs).
- ➔ **Fraud Detection:** Detecting fraudulent transactions in banking or e-commerce.

Thanks!

Any
questions?